

RAG Guide

1. What is RAG?

Retrieval-Augmented Generation (RAG) is a technique that improves the responses generated by Large Language Models by providing relevant information from an external knowledge source.

2. Why is RAG Needed?

Language models may not contain the specific or up-to-date information required for a task. RAG allows an application to retrieve relevant information from a knowledge base before generating an answer.

3. How RAG Works

A typical RAG pipeline has two major stages: retrieval and generation. First, the user's query is used to retrieve relevant documents or chunks. Then, the retrieved context is provided to an LLM so it can generate an answer.

4. Embeddings

Embeddings represent text as numerical vectors. Texts with similar meanings tend to have vectors that are close to each other in vector space. Embeddings are commonly used to perform semantic search.

5. Vector Databases

A vector database stores embeddings and supports similarity search. Examples include ChromaDB and Pinecone. A retrieval system can use these databases to find chunks that are semantically relevant to a user's query.

6. Retrieval

During retrieval, the system searches the knowledge base for information relevant to the user's query. The retrieved chunks become context for the generation step.

7. Generation

The retrieved context is passed to a Large Language Model. The model uses that context to generate a response that is grounded in the available information.

8. Advantages of RAG

RAG can connect an LLM to private or domain-specific information, reduce the need to put all information directly into a prompt, and make it possible to update the knowledge source without retraining the model.

9. Limitations of RAG

RAG quality depends on retrieval quality. If the relevant information is not retrieved, the LLM may not have the context needed to answer correctly. Chunking, embeddings, retrieval strategy, and ranking therefore affect the final result.